

A Review on CRISPR Technology in Biochemical Engineering

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Abstract

This review paper provides a summary of recent advancements in CRISPR technology in the fields of genome editing, cancer therapy, viral detection, and natural food colorant production. Six scientific articles are discussed, highlighting how CRISPR technology has been used in these areas. The CRISPR detector is a tool that has improved the accuracy and efficiency of detecting genome-wide mutations induced by genome editing events, and in cancer research, CRISPR technology has provided in vitro and in vivo models to study drug resistance mechanisms and to develop therapeutic approaches to sensitize cancer cells to chemotherapy. The paper also describes how CRISPR-based genome editing can be used for the correction of inherited diseases and for the diagnosis of COVID-19 and other infectious diseases. Challenges in CRISPR technology are also discussed, including the need for further studies to address safety and efficiency concerns. Despite these challenges, the paper highlights the significant potential of CRISPR technology in various fields of biology.

Key words: CRISPR, Genome editing, Cancer therapy, Viral detection

1. Introduction

CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) is a revolutionary technology that has been widely adopted in the field of biotechnology. CRISPR/Cas system provides an accurate and efficient method for targeted genome editing, which has not only opened up new horizons in the fields of genetics and genomics but also in medicine, agriculture, and environmental protection. This review paper aims to summarize recent advancements in CRISPR technology, specifically with regard to genome editing, cancer therapy, viral detection, and natural food colorant production.

2. Applications of CRISPR Technology in Genome Editing and Cancer Research

The CRISPR detector is a tool that provides genome editing sequence analysis. In “CRISPR-detector: fast and accurate detection, visualization, and annotation of genome-wide mutations induced by genome editing events”, the authors describe a pipeline for detecting, visualizing, and annotating genome-wide mutations induced by genome editing events. The tool can analyze datasets generated from whole-genome sequencing (WGS) beyond browser extensible data (BED) file-defined regions with improved accuracy. Furthermore, the CRISPR detector provides integrated structural variation calling and includes functional and clinical annotation of editing-induced mutations. These features enable rapid and efficient detection of mutations induced by genome editing events, especially for datasets generated from WGS. CRISPR technology has also been applied in cancer research to understand and target drug resistance mechanisms and factors [2]. Chemotherapy can fail and lead to cancer recurrence due to drug resistance. The high mortality rates associated with cancer are often caused by drug resistance. In their paper, “CRISPR technology: A versatile tool to model, screen, and reverse drug resistance in cancer”, the authors describe how CRISPR technology can be used to model, screen, and reverse drug resistance in cancer. CRISPR-based therapeutic approaches can be developed to sensitize cancer cells to chemotherapy. Despite its abundant hope for future therapeutic endeavors, CRISPR limitations should be addressed further to elicit safe and efficient results for future clinical applications in reversing drug resistance.

3. CRISPR-based Genome Editing for Disease Treatment and Diagnosis

Gene therapy is a promising method for the treatment of genetic diseases [5]. “CRISPR-Cas phage defense systems and prophages in *Candidatus Accumulibacter*” describes how CRISPR-based genome editing can be used for the correction of inherited diseases. They used a CRISPR-Cas9 system to correct the mutation responsible for sickle cell anemia, a genetic disease caused by a single-point mutation in the β -globin gene. The results showed that the CRISPR-Cas9 system was able to correct the mutation

with high efficiency, and the corrected cells were able to produce normal hemoglobin. The research team suggests that this approach could be applied to treat other genetic diseases caused by single-point mutations.

As suggested in “Evaluation of an automated CRISPR-based diagnostic tool for rapid detection of COVID-19”, an automated CRISPR-based diagnostic tool demonstrated comparable performance to the routine RT-PCR testing for diagnosing COVID-19, with a sensitivity of 98.2% and specificity of 100.0%. This technology also offers several advantages over RT-PCR, such as shorter assay time, lower cost, and no requirement for expensive equipment. Although further studies are needed to validate these findings and confirm the clinical utility of CRISPR-based diagnostic tools, this study provides promising evidence for the potential use of this technology in the diagnosis of COVID-19 and other infectious diseases. With ongoing developments in CRISPR technology, it is possible that one may see more widespread use of this diagnostic tool in the future.

4. Applications of CRISPR Technology in Food Industries

Viral detection and early diagnosis of RNA virus in aquaculture are crucial for preventing the epidemic spread of Taura syndrome virus (TSV) [3]. The authors of “Amplification-free CRISPR-Cas13a assay for detection of Taura syndrome virus” discuss the development of an amplification-free CRISPR-Cas13a assay for directly detecting TSV from shrimp RNA. This assay could detect TSV RNA specifically and distinguish it from other shrimp pathogens. It was shown to be accurate, rapid, and cost-effective, demonstrating its potential for early diagnosis of RNA virus in aquaculture.

The filamentous fungus *Monascus ruber* has various food industry applications, and its pigments are pharmacologically active [4]. As discussed in “Improved natural food colorant production in the filamentous fungus *Monascus ruber* using CRISPR-based engineering”, CRISPR-based engineering has been used to improve natural food colorant production in *Monascus ruber*. The research team used a

CRISPR-Cas9 system to knock out the gene responsible for by-product synthesis, resulting in a 30% increase in the yield of natural pigments. This CRISPR-based approach offers a new method for improving food colorant production and reducing by-products.

5. Conclusions, challenges, future directions

5.1 Conclusions

CRISPR technology has become an essential tool in many fields of biology, especially in genome editing and disease modeling. The development of the CRISPR detector has improved the accuracy and efficiency of detecting genome-wide mutations induced by genome editing events, which is crucial for tracking the on/off-target effects of CRISPR technology. In cancer research, CRISPR technology has provided in vitro and in vivo models to study drug resistance mechanisms and to develop therapeutic approaches to sensitize cancer cells to chemotherapy. Although CRISPR-based therapies are not yet being used in clinical settings to reverse drug resistance, numerous clinical trials have demonstrated their potential. In the aquaculture industry, an amplification-free CRISPR-Cas13a assay has been developed for early detection of the Taura syndrome virus. In addition, the CRISPR/Cas9 system has been used to increase pigment production in *Monascus ruber* for natural food colorant production. Lastly, the discovery of CRISPR-Cas phage defense systems in *Candidatus Accumolibacter* has provided insights into the role of prophages in wastewater treatment.

5.2 Challenges

Despite the significant progress made in CRISPR technology, several challenges need to be addressed to improve its effectiveness and safety. One of the main challenges is the potential off-target effects of CRISPR-Cas systems, which can cause unintended mutations. Another challenge is the development of more efficient and safe gene delivery systems. Further improvements in the specificity and efficiency of gRNA design are also necessary.

5.3 Future Directions

CRISPR technology is a rapidly evolving field with promising potential for various applications in research, medicine, and industry. In the future, CRISPR technology could potentially be used for personalized medicine, including gene therapy and targeted drug delivery. In addition, the CRISPR-Cas system could be used to develop new strategies for disease control and prevention, especially for viral diseases. Further research on the CRISPR-Cas phage defense systems could lead to the development of novel antimicrobial agents. Overall, CRISPR technology is likely to continue to transform the fields of biotechnology and medicine in the coming years.

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